



Contained Availability Groups

- Best practices and real-world challenges

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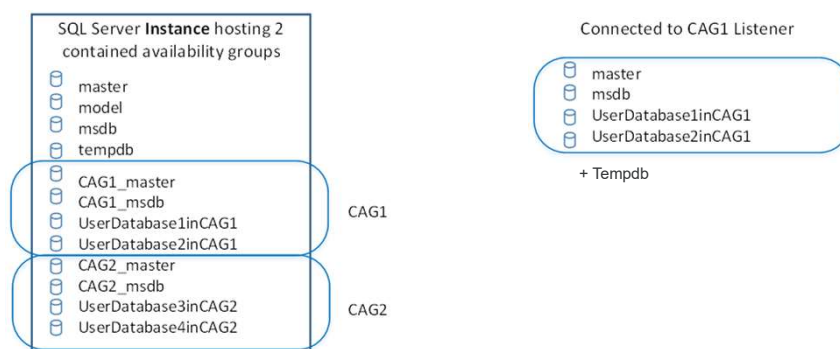
Goals



- The good, the bad, the ugly: What differentiates cAGs
 - Availability
 - Security
 - Management
 - Deployment
- Best practices

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Quick memory refresh on how Contained Availability Group present themselves



Note: Contained Availability Groups can not be used without Listener

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AVAILABILITY

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Availability

- No changes in HA itself
- But includes all the dependencies
 - Logins
 - ▶ Server permissions
 - Certificates
 - Jobs
 - ▶ Schedules
 - ▶ Proxies

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SECURITY

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Security

- Sysadmins will be copied at CAG creation
- Roles
 - Most of the new server roles are not available inside CAGs
- Audits
 - Per cAG
 - Naming needs to differ

Keep in mind for security assessments!

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MANAGEMENT

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Jobs

- Jobs always run on the primary only
- You need to decide your strategy
 - Which jobs at which scope
 - ▶ Does 3rd level support need to see maintenance jobs?
- Backup
 - Backup history is in CAG – get script from there
 - Restore history on host!
 - ▶ Affects reports
 - Transaction markers are on host!

Always be aware of your connection-context!

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DEPLOYMENT

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Deployment

- In an ideal world, everything would happen within the AG
 - Except deployment of new databases
 - ▶ 2 steps due to connection change
 - ▷ DBs create
 - ▷ then move to cAG
 - ▶ What other server level objects are deployed?

Always be aware of your connection-context!

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Limits and differences

- Server config only from host master
- A plus: list of databases limited
- 3rd party support lacking

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Biggest Challenges

- Get a complete list of accounts with access to SQL Server
- Keeping oversight of scheduled jobs to avoid bottlenecks
- Database-deployment still has to start on the host
 - Feedback "Allow creation and restore of databases in contained availability group" for MS: <https://feedback.azure.com/d365community/idea/07f0807c-6825-f011-9d47-7c1e52d4bdd3>
- No simple way to enumerate all cAG connections
- Easy to confuse a cAG-connectin with a real server – no trivial flags to check are available

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Best Practices



- Always be aware of your environment (cAG vs Server)
 - AG Listener separate from Hosts
- Split critical Apps
- Use DNS Alias
- Check your tools

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Known Issues



- Backup history is inside AG, but transaction (WITH MARK) markers are saved in the host msdb logmarkhistory-table
 - After failover on a different host even
 - ▶ Restore with STOPATMARK/STOPBEFOREMARK requires the knowledge of the mark
 - ▶ [Contained Availability Group writes transaction marks to host instead of contained msdb : Community](#)
 - ▶ (Restore statement is generated inside the cAG but run on the host)
- Jobs cannot log to table when job part of Contained Availability Group
 - [Jobs cannot log to table when job part of Contained Availability Group : Community](#)
- Cannot use same Audit name as on Server unless GUID is the same
 - <https://feedback.azure.com/d365community/idea/1358762b-64e3-ef11-b542-00224854717c>
- Bug: contained master database shows page_verify option set to NONE plus it cannot be changed (started with CU16)
 - <https://feedback.azure.com/d365community/idea/cf69002b-53ea-ef11-a4de-0022484ba63a>
- Bug: contained msdb has Trustworthy bit set OFF (in Contained Availability Group)
 - [Bug: contained msdb has Trustworthy bit set OFF \(in Contained Availability Group\) : Community](#)

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Unsupported functionalities in cAGs

- SSIS packages
- SQL Server Replication
- (cAG as Logshipping target – true for any type of AG)

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Just a few resources on CAGs

- Why you should use SQL Server contained availability groups to save time – and why consultants may not tell you about them
 - https://andreas-wolter.com/en/2504_sqlserver_contained_availability_groups/
- What is a contained availability group?
 - <https://learn.microsoft.com/en-us/sql/database-engine/availability-groups/windows/contained-availability-groups-overview?view=sql-server-ver16>

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Thank you!



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